# AERIS Expert Review Packet

## Anomaly

| | |
|---|---|
| Source / metric | tceq / Carbon monoxide (co) |
| Observed / expected baseline | 0.5 / 0.155725 ppm |
| Baseline method | mean of the preceding 7 days of this series (Z-score detector) |
| Time | 2026-07-10 16:00 UTC |
| Location | 29.6697, -95.1285 |
| Severity | severe |
| Detectors | zscore, stl, isolation_forest |

## Stored evidence summary

The three tables in this section are look-up tables. You never have to read them through: they are here for the moment a claim quotes a number and you want to check it.

| Source | Metric | Unit | Entities / points | Range; mean | Nearest to event |
|---|---|---|---|---|---|
| asos | Relative humidity (humidity) | percent | 7 / 724 | 43.23 to 100; mean 75.9835 | 66.44 at 2026-07-10 15:55Z; 5 min before event |
| asos | Air temperature (temperature) | degC | 7 / 724 | 23.6111 to 35.5556; mean 28.5678 | 31 at 2026-07-10 15:55Z; 5 min before event |
| asos | Wind direction (wind_direction) | degree | 7 / 896 | 0 to 360; mean 125.659 | 200 at 2026-07-10 15:55Z; 5 min before event |
| asos | Wind speed (wind_speed) | m/s | 7 / 918 | 0 to 10.8033; mean 2.5308 | 2.57222 at 2026-07-10 15:55Z; 5 min before event |
| noaa_gfs | 500-hPa geopotential height (gh_500) | m | 8 / 96 | 5905.85 to 5931.15; mean 5919.26 | 5920.27 at 2026-07-10 18:00Z; 2 h after event |
| noaa_gfs | Planetary boundary layer height (pbl_height) | m | 8 / 96 | 12.0586 to 1950.8; mean 659.161 | 1686.16 at 2026-07-10 18:00Z; 2 h after event |
| noaa_gfs | Precipitable water (precipitable_water) | mm | 8 / 96 | 43.0571 to 58.1093; mean 50.9795 | 55.9306 at 2026-07-10 18:00Z; 2 h after event |
| noaa_gfs | Surface pressure (surface_pressure) | hPa | 8 / 96 | 1011.3 to 1017.46; mean 1014.15 | 1014.56 at 2026-07-10 18:00Z; 2 h after event |
| noaa_gfs | 850-hPa temperature (t_850) | degC | 8 / 96 | 17.2076 to 20.7322; mean 18.8377 | 17.2076 at 2026-07-10 18:00Z; 2 h after event |
| noaa_gfs | 10-m eastward wind (u_10m) | m/s | 8 / 96 | -3.47027 to 1.76546; mean -0.7381 | -2.37843 at 2026-07-10 18:00Z; 2 h after event |
| noaa_gfs | 10-m northward wind (v_10m) | m/s | 8 / 96 | -0.626672 to 5.52466; mean 2.283 | 2.66054 at 2026-07-10 18:00Z; 2 h after event |
| openaq | Ozone (ozone) | ppm | 17 / 950 | -0.003 to 0.054; mean 0.0164 | 0.023 at 2026-07-10 16:00Z; at event time |

| Source | Metric | Unit | Entities / points | Range; mean | Nearest to event |
|---|---|---|---|---|---|
| openaq | PM10 (pm10) | ug/m3 | 2 / 119 | 9 to 60; mean 27.9664 | 24 at 2026-07-10 16:00Z; at event time |
| openaq | PM2.5 (pm25) | ug/m3 | 77 / 3547 | 0 to 25.9; mean 5.3449 | 9.9 at 2026-07-10 16:00Z; at event time |
| openweather | Cloud cover (cloud_cover) | percent | 5 / 360 | 0 to 100; mean 49.8444 | 79 at 2026-07-10 16:01Z; 2 min after event |
| openweather | Relative humidity (humidity) | percent | 5 / 360 | 52 to 96; mean 78.6556 | 80 at 2026-07-10 16:01Z; 2 min after event |
| openweather | Precipitation rate (precipitation) | mm/h | 5 / 360 | 0 to 7.7; mean 0.0798 | 0 at 2026-07-10 16:01Z; 2 min after event |
| openweather | Surface pressure (pressure) | hPa | 5 / 360 | 1014 to 1018; mean 1016.06 | 1017 at 2026-07-10 16:01Z; 2 min after event |
| openweather | Air temperature (temperature) | degC | 5 / 360 | 25.01 to 35.63; mean 29.0433 | 30.26 at 2026-07-10 16:01Z; 2 min after event |
| openweather | Wind direction (wind_direction) | degree | 5 / 360 | 0 to 333; mean 165.558 | 161 at 2026-07-10 16:01Z; 2 min after event |
| openweather | Wind speed (wind_speed) | m/s | 5 / 360 | 0.45 to 6.21; mean 2.3174 | 3.41 at 2026-07-10 16:01Z; 2 min after event |
| purpleair | PM2.5 (pm25) | ug/m3 | 65 / 4458 | 0 to 102.8; mean 6.8215 | 7.7 at 2026-07-10 15:59Z; at event time |
| sentinel5p | Aerosol index granules (s5p_aer_ai_granule_available) | count | 6 / 6 | 1 to 1; mean 1 | 1 at 2026-07-10 19:11Z; 3 h 11 min after event |
| sentinel5p | CH4 granules available (s5p_ch4_granule_available) | count | 5 / 5 | 1 to 1; mean 1 | 1 at 2026-07-10 19:11Z; 3 h 11 min after event |
| sentinel5p | CO column (s5p_co_column) | mol/m^2 | 4 / 4 | 0.0242691 to 0.0262775; mean 0.0253 | 0.0262775 at 2026-07-10 19:11Z; 3 h 11 min after event |
| sentinel5p | CO granules available (s5p_co_granule_available) | count | 6 / 6 | 1 to 1; mean 1 | 1 at 2026-07-10 19:11Z; 3 h 11 min after event |
| sentinel5p | HCHO column (s5p_hcho_column) | mol/m^2 | 4 / 4 | 0.000148478 to 0.000160047; mean 0.0002 | 0.000160047 at 2026-07-10 19:11Z; 3 h 11 min after event |
| sentinel5p | HCHO granules available (s5p_hcho_granule_available) | count | 6 / 6 | 1 to 1; mean 1 | 1 at 2026-07-10 19:11Z; 3 h 11 min after event |
| sentinel5p | NO2 column (s5p_no2_column) | mol/m^2 | 2 / 2 | 3.3321e-05 to 4.40738e-05; mean 0 | 4.40738e-05 at 2026-07-10 19:39Z; 3 h 39 min after event |
| sentinel5p | NO2 granules available (s5p_no2_granule_available) | count | 3 / 3 | 1 to 1; mean 1 | 1 at 2026-07-10 19:39Z; 3 h 39 min after event |
| sentinel5p | O3 granules available (s5p_o3_granule_available) | count | 6 / 6 | 1 to 1; mean 1 | 1 at 2026-07-10 19:11Z; 3 h 11 min after event |
| sentinel5p | SO2 column (s5p_so2_column) | mol/m^2 | 4 / 4 | -5.8828e-06 to 2.37917e-05; mean 0 | 2.37917e-05 at 2026-07-10 19:11Z; 3 h 11 min after event |
| sentinel5p | SO2 granules available (s5p_so2_granule_available) | count | 6 / 6 | 1 to 1; mean 1 | 1 at 2026-07-10 19:11Z; 3 h 11 min after event |
| tceq | Carbon monoxide (co) | ppm | 3 / 191 | 0.1 to 0.9; mean 0.2445 | 0.5 at 2026-07-10 16:00Z; at event time |
| tceq | Nitrogen dioxide (no2) | ppb | 12 / 745 | -1.9 to 20.3; mean 5.4154 | 1.9 at 2026-07-10 16:00Z; at event time |

| Source | Metric | Unit | Entities / points | Range; mean | Nearest to event |
|---|---|---|---|---|---|
| tceq | Sulfur dioxide (so2) | ppb | 3 / 185 | -0.6 to 3.7; mean -0.0978 | 0.2 at 2026-07-10 16:00Z; at event time |

## Six-hour averages

Each metric averaged in 6-hour blocks across the stored 72 hours, in UTC. These averages, and the per-site ones below, are what the models were shown.

| Source | Metric | Values |
|---|---|---|
| asos | Relative humidity (humidity) | 07-09 00Z 82.72, 06Z 88.61, 12Z 67.82, 18Z 56.47, 07-10 00Z 75.84, 06Z 87.02, 12Z 74.65, 18Z 62.43, 07-11 00Z 78.3, 06Z 88.71, 12Z 79.97, 18Z 70.7, 07-12 00Z 85.94 |
| asos | Air temperature (temperature) | 07-09 00Z 26.99, 06Z 25.75, 12Z 30.42, 18Z 32.36, 07-10 00Z 28.37, 06Z 26.29, 12Z 29.18, 18Z 31.39, 07-11 00Z 27.77, 06Z 26.07, 12Z 27.89, 18Z 29.44, 07-12 00Z 27.19 |
| asos | Wind direction (wind_direction) | 07-09 00Z 143.5, 06Z 121.7, 12Z 163.3, 18Z 171.7, 07-10 00Z 146.1, 06Z 80.67, 12Z 118.7, 18Z 156.7, 07-11 00Z 135.7, 06Z 64.13, 12Z 131.6, 18Z 108.1, 07-12 00Z 91.6 |
| asos | Wind speed (wind_speed) | 07-09 00Z 2.196, 06Z 1.418, 12Z 2.376, 18Z 5.23, 07-10 00Z 2.779, 06Z 1.128, 12Z 2.412, 18Z 4.476, 07-11 00Z 2.281, 06Z 0.8867, 12Z 2.258, 18Z 2.762, 07-12 00Z 2.315 |
| noaa_gfs | 500-hPa geopotential height (gh_500) | 07-09 06Z 5919, 12Z 5910, 18Z 5921, 07-10 00Z 5915, 06Z 5921, 12Z 5907, 18Z 5920, 07-11 00Z 5919, 06Z 5924, 12Z 5919, 18Z 5930, 07-12 00Z 5927 |
| noaa_gfs | Planetary boundary layer height (pbl_height) | 07-09 06Z 291.9, 12Z 144.3, 18Z 1731, 07-10 00Z 883.8, 06Z 203.7, 12Z 115.4, 18Z 1552, 07-11 00Z 975.9, 06Z 280.3, 12Z 70.06, 18Z 854.7, 07-12 00Z 807.6 |
| noaa_gfs | Precipitable water (precipitable_water) | 07-09 06Z 44.47, 12Z 44.38, 18Z 45.86, 07-10 00Z 46.68, 06Z 46.47, 12Z 53.7, 18Z 55.67, 07-11 00Z 53.48, 06Z 52.82, 12Z 55.97, 18Z 55.83, 07-12 00Z 56.43 |
| noaa_gfs | Surface pressure (surface_pressure) | 07-09 06Z 1015, 12Z 1015, 18Z 1015, 07-10 00Z 1013, 06Z 1013, 12Z 1014, 18Z 1014, 07-11 00Z 1013, 06Z 1014, 12Z 1015, 18Z 1016, 07-12 00Z 1014 |
| noaa_gfs | 850-hPa temperature (t_850) | 07-09 06Z 19.44, 12Z 19.05, 18Z 17.77, 07-10 00Z 18.86, 06Z 20.5, 12Z 18.53, 18Z 17.67, 07-11 00Z 18.52, 06Z 19.37, 12Z 18.76, 18Z 18.34, 07-12 00Z 19.26 |
| noaa_gfs | 10-m eastward wind (u_10m) | 07-09 06Z 0.9761, 12Z 1.052, 18Z -0.326, 07-10 00Z -0.9792, 06Z -0.937, 12Z 0.1127, 18Z -2.221, 07-11 00Z -1.981, 06Z -1.098, 12Z -0.3907, 18Z -0.4401, 07-12 00Z -2.624 |
| noaa_gfs | 10-m northward wind (v_10m) | 07-09 06Z 2.668, 12Z 1.384, 18Z 2.518, 07-10 00Z 4.73, 06Z 2.002, 12Z 1.276, 18Z 3.214, 07-11 00Z 3.62, 06Z 2.315, 12Z 0.5108, 18Z 1.001, 07-12 00Z 2.157 |
| openaq | Ozone (ozone) | 07-09 12Z 0.01656, 18Z 0.0252, 07-10 00Z 0.01537, 06Z 0.006988, 12Z 0.01161, 18Z 0.02268, 07-11 00Z 0.01203, 06Z 0.006133, 12Z 0.01238, 18Z 0.02849, 07-12 00Z 0.01893 |
| openaq | PM10 (pm10) | 07-09 12Z 27.25, 18Z 26.75, 07-10 00Z 15.5, 06Z 18.83, 12Z 28.33, 18Z 35.27, 07-11 00Z 42.45, 06Z 33.08, 12Z 34.27, 18Z 23.17, 07-12 00Z 23.89 |
| openaq | PM2.5 (pm25) | 07-09 00Z 4.437, 06Z 4.489, 12Z 4.986, 18Z 4.063, 07-10 00Z 3.709, 06Z 5.629, 12Z 6.406, 18Z 5.188, 07-11 00Z 6.032, 06Z 6.073, 12Z 5.549, 18Z 5.589, 07-12 00Z 5.505 |
| openweather | Cloud cover (cloud_cover) | 07-09 00Z 49.2, 06Z 31.8, 12Z 3.733, 18Z 23.33, 07-10 00Z 88.62, 06Z 47.74, 12Z 84.67, 18Z 87.21, 07-11 00Z 25, 06Z 11.03, 12Z 73.84, 18Z 88.41, 07-12 00Z 29.48 |
| openweather | Relative humidity (humidity) | 07-09 00Z 79.9, 06Z 88.2, 12Z 76.87, 18Z 60.7, 07-10 00Z 74.24, 06Z 88.13, 12Z 79.8, 18Z 66.86, 07-11 00Z 76.9, 06Z 87.79, 12Z 86.58, 18Z 74, 07-12 00Z 84.1 |
| openweather | Precipitation rate (precipitation) | 07-09 00Z 0, 06Z 0, 12Z 0, 18Z 0.2567, 07-10 00Z 0, 06Z 0, 12Z 0.1113, 18Z 0.003448, 07-11 00Z 0, 06Z 0, 12Z 0.3806, 18Z 0.1293, 07-12 00Z 0.09762 |
| openweather | Surface pressure (pressure) | 07-09 00Z 1017, 06Z 1016, 12Z 1018, 18Z 1015, 07-10 00Z 1015, 06Z 1015, 12Z 1017, 18Z 1015, 07-11 00Z 1015, 06Z 1016, 12Z 1018, 18Z 1016, 07-12 00Z 1016 |
| openweather | Air temperature (temperature) | 07-09 00Z 28.01, 06Z 26.28, 12Z 29.95, 18Z 34.18, 07-10 00Z 29.64, 06Z 26.49, 12Z 29.45, 18Z 32.61, 07-11 00Z 28.81, 06Z 26.15, 12Z 27.24, 18Z 30.08, 07-12 00Z 27.85 |
| openweather | Wind direction (wind_direction) | 07-09 00Z 160.8, 06Z 172.6, 12Z 216.1, 18Z 179.6, 07-10 00Z 182.1, 06Z 171.2, 12Z 171.4, 18Z 150.1, 07-11 00Z 146, 06Z 142.9, 12Z 162.5, 18Z 145, 07-12 00Z 140.6 |
| openweather | Wind speed (wind_speed) | 07-09 00Z 2.706, 06Z 1.932, 12Z 1.636, 18Z 2.793, 07-10 00Z 2.224, 06Z 1.664, 12Z 1.934, 18Z 3.691, 07-11 00Z 1.956, 06Z 1.898, 12Z 2.659, 18Z 3.01, 07-12 00Z 2.373 |

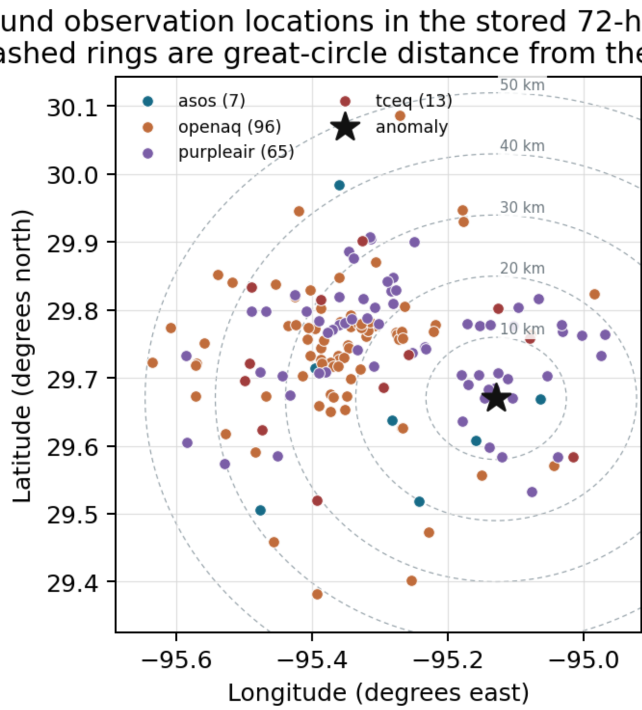
| Source | Metric | Values |
|---|---|---|
| purpleair | PM2.5 (pm25) | 07-09 00Z 5.683, 06Z 6.844, 12Z 6.313, 18Z 3.945, 07-10 00Z 5.904, 06Z 7.399, 12Z 7.635, 18Z 6.472, 07-11 00Z 8.086, 06Z 8.661, 12Z 7.124, 18Z 6.638, 07-12 00Z 7.495 |
| sentinel5p | Aerosol index granules (s5p_aer_ai_granule_available) | 07-09 18Z 1, 07-10 18Z 1, 07-11 18Z 1 |
| sentinel5p | CH4 granules available (s5p_ch4_granule_available) | 07-09 18Z 1, 07-10 18Z 1, 07-11 18Z 1 |
| sentinel5p | CO column (s5p_co_column) | 07-09 18Z 0.02438, 07-10 18Z 0.02624 |
| sentinel5p | CO granules available (s5p_co_granule_available) | 07-09 18Z 1, 07-10 18Z 1, 07-11 18Z 1 |
| sentinel5p | HCHO column (s5p_hcho_column) | 07-09 18Z 0.0001505, 07-10 18Z 0.0001571 |
| sentinel5p | HCHO granules available (s5p_hcho_granule_available) | 07-09 18Z 1, 07-10 18Z 1, 07-11 18Z 1 |
| sentinel5p | NO2 column (s5p_no2_column) | 07-09 18Z 3.332e-05, 07-10 18Z 4.407e-05 |
| sentinel5p | NO2 granules available (s5p_no2_granule_available) | 07-09 18Z 1, 07-10 18Z 1, 07-11 18Z 1 |
| sentinel5p | O3 granules available (s5p_o3_granule_available) | 07-09 18Z 1, 07-10 18Z 1, 07-11 18Z 1 |
| sentinel5p | SO2 column (s5p_so2_column) | 07-09 18Z 2.306e-05, 07-10 18Z 8.954e-06 |
| sentinel5p | SO2 granules available (s5p_so2_granule_available) | 07-09 18Z 1, 07-10 18Z 1, 07-11 18Z 1 |
| tceq | Carbon monoxide (co) | 07-09 06Z 0.2222, 12Z 0.2722, 18Z 0.1667, 07-10 00Z 0.1667, 06Z 0.2167, 12Z 0.2722, 18Z 0.2278, 07-11 00Z 0.21, 06Z 0.2389, 12Z 0.3167, 18Z 0.2389, 07-12 00Z 0.4 |
| tceq | Nitrogen dioxide (no2) | 07-09 06Z 5.834, 12Z 5.389, 18Z 3.253, 07-10 00Z 2.408, 06Z 5.294, 12Z 6.969, 18Z 5.342, 07-11 00Z 4.884, 06Z 6.561, 12Z 5.349, 18Z 5.765, 07-12 00Z 6.958 |
| tceq | Sulfur dioxide (so2) | 07-09 06Z -0.1556, 12Z -0.09412, 18Z -0.1667, 07-10 00Z -0.1444, 06Z -0.2056, 12Z -0.1833, 18Z -0.1889, 07-11 00Z -0.12, 06Z -0.1944, 12Z -0.1833, 18Z 0.5389, 07-12 00Z -0.14 |

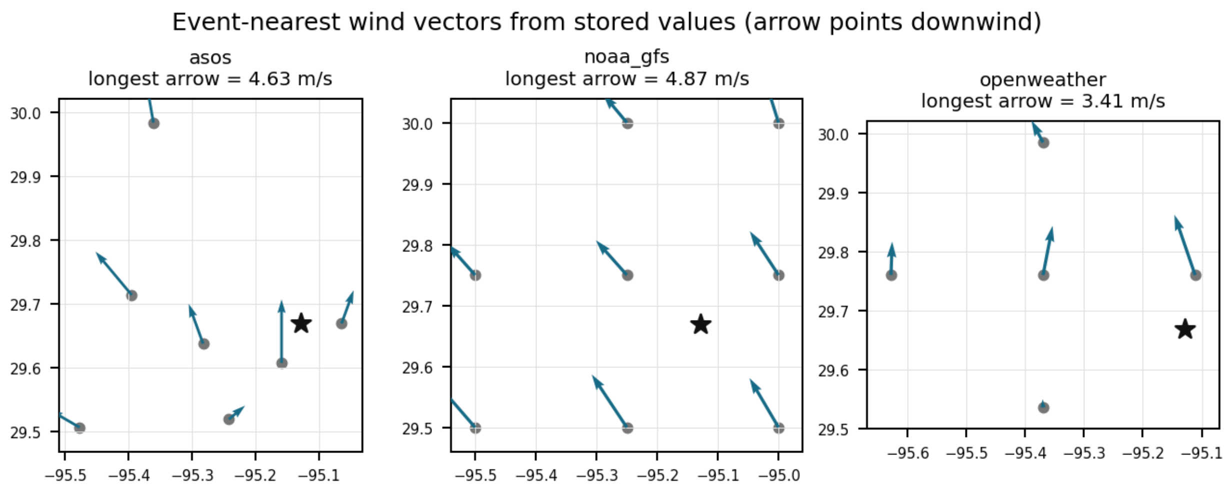
## Per-site averages

Each metric averaged at each site, with that site's distance from the event in brackets.

| Source | Metric | Values |
|---|---|---|
| asos | Relative humidity (humidity) | T41 (6.225 km) 77.2, EFD (7.53 km) 74.79, HOU (15.295 km) 75.57, LVJ (20.026 km) 74.49, MCJ (26.207 km) 75.26, AXH (38.286 km) 76.84, IAH (41.543 km) 76.37 |
| asos | Air temperature (temperature) | T41 (6.225 km) 28.77, EFD (7.53 km) 29, HOU (15.295 km) 28.74, LVJ (20.026 km) 28.8, MCJ (26.207 km) 28.7, AXH (38.286 km) 27.72, IAH (41.543 km) 28.78 |
| asos | Wind direction (wind_direction) | T41 (6.225 km) 148.9, EFD (7.53 km) 125.8, HOU (15.295 km) 144.6, LVJ (20.026 km) 100.6, MCJ (26.207 km) 151.1, AXH (38.286 km) 75.24, IAH (41.543 km) 137.4 |
| asos | Wind speed (wind_speed) | T41 (6.225 km) 2.822, EFD (7.53 km) 3.132, HOU (15.295 km) 2.879, LVJ (20.026 km) 1.901, MCJ (26.207 km) 3.655, AXH (38.286 km) 1.043, IAH (41.543 km) 2.651 |
| noaa_gfs | 500-hPa geopotential height (gh_500) | gfs:29.75,-95.25 (14.741 km) 5919, gfs:29.75,-95.00 (15.288 km) 5920, gfs:29.50,-95.25 (22.231 km) 5919, gfs:29.50,-95.00 (22.598 km) 5920, gfs:29.75,-95.50 (36.97 km) 5919, gfs:30.00,-95.25 (38.548 km) 5919, gfs:30.00,-95.00 (38.76 km) 5920, gfs:29.50,-95.50 (40.577 km) 5919 |
| noaa_gfs | Planetary boundary layer height (pbl_height) | gfs:29.75,-95.25 (14.741 km) 762, gfs:29.75,-95.00 (15.288 km) 727.3, gfs:29.50,-95.25 (22.231 km) 457, gfs:29.50,-95.00 (22.598 km) 585.9, gfs:29.75,-95.50 (36.97 km) 792, gfs:30.00,-95.25 (38.548 km) 838.9, gfs:30.00,-95.00 (38.76 km) 545.4, gfs:29.50,-95.50 (40.577 km) 564.7 |

| Source | Metric | Values |
|---|---|---|
| noaa_gfs | Precipitable water (precipitable_water) | gfs:29.75,-95.25 (14.741 km) 50.77, gfs:29.75,-95.00 (15.288 km) 51.12, gfs:29.50,-95.25 (22.231 km) 51.45, gfs:29.50,-95.00 (22.598 km) 51.83, gfs:29.75,-95.50 (36.97 km) 50.74, gfs:30.00,-95.25 (38.548 km) 50.32, gfs:30.00,-95.00 (38.76 km) 50.55, gfs:29.50,-95.50 (40.577 km) 51.04 |
| noaa_gfs | Surface pressure (surface_pressure) | gfs:29.75,-95.25 (14.741 km) 1014, gfs:29.75,-95.00 (15.288 km) 1015, gfs:29.50,-95.25 (22.231 km) 1015, gfs:29.50,-95.00 (22.598 km) 1016, gfs:29.75,-95.50 (36.97 km) 1013, gfs:30.00,-95.25 (38.548 km) 1013, gfs:30.00,-95.00 (38.76 km) 1014, gfs:29.50,-95.50 (40.577 km) 1014 |
| noaa_gfs | 850-hPa temperature (t_850) | gfs:29.75,-95.25 (14.741 km) 18.82, gfs:29.75,-95.00 (15.288 km) 18.87, gfs:29.50,-95.25 (22.231 km) 18.82, gfs:29.50,-95.00 (22.598 km) 18.86, gfs:29.75,-95.50 (36.97 km) 18.75, gfs:30.00,-95.25 (38.548 km) 18.86, gfs:30.00,-95.00 (38.76 km) 18.88, gfs:29.50,-95.50 (40.577 km) 18.83 |
| noaa_gfs | 10-m eastward wind (u_10m) | gfs:29.75,-95.25 (14.741 km) -0.4465, gfs:29.75,-95.00 (15.288 km) -0.6573, gfs:29.50,-95.25 (22.231 km) -0.7732, gfs:29.50,-95.00 (22.598 km) -0.6898, gfs:29.75,-95.50 (36.97 km) -0.8248, gfs:30.00,-95.25 (38.548 km) -0.6923, gfs:30.00,-95.00 (38.76 km) -0.7207, gfs:29.50,-95.50 (40.577 km) -1.1 |
| noaa_gfs | 10-m northward wind (v_10m) | gfs:29.75,-95.25 (14.741 km) 2.325, gfs:29.75,-95.00 (15.288 km) 2.295, gfs:29.50,-95.25 (22.231 km) 2.213, gfs:29.50,-95.00 (22.598 km) 2.727, gfs:29.75,-95.50 (36.97 km) 2.317, gfs:30.00,-95.25 (38.548 km) 2.033, gfs:30.00,-95.00 (38.76 km) 1.957, gfs:29.50,-95.50 (40.577 km) 2.397 |
| openaq | Ozone (ozone) | 332 (0.032 km) 0.021, 2800 (11.598 km) 0.01768, 2039 (14.096 km) 0.01569, 320 (14.272 km) 0.01437, 3378 (14.356 km) 0.01327, 5077791 (14.558 km) 0.01547, 3214 (14.765 km) 0.01603, 325 (16.163 km) 0.004614, +9 more |
| openaq | PM10 (pm10) | 271 (0.032 km) 23.33, 15852419 (21.358 km) 32.68 |
| openaq | PM2.5 (pm25) | 268 (0.032 km) 9.193, 13955815 (12.727 km) 5.355, 14404954 (13.725 km) 2.613, 14157975 (14.275 km) 8.603, 15539682 (14.359 km) 6.969, 8967123 (14.413 km) 7.731, 8967479 (14.413 km) 7.161, 13787195 (14.82 km) 7.406, +69 more |
| openweather | Cloud cover (cloud_cover) | grid:east (10.221 km) 51.78, grid:center (25.388 km) 48.85, grid:south (27.676 km) 51.62, grid:north (42.079 km) 51.93, grid:west (49.325 km) 45.04 |
| openweather | Relative humidity (humidity) | grid:east (10.221 km) 81.99, grid:center (25.388 km) 78.29, grid:south (27.676 km) 77.64, grid:north (42.079 km) 77.51, grid:west (49.325 km) 77.85 |
| openweather | Precipitation rate (precipitation) | grid:east (10.221 km) 0.1008, grid:center (25.388 km) 0.08833, grid:south (27.676 km) 0.07028, grid:north (42.079 km) 0.1371, grid:west (49.325 km) 0.002639 |
| openweather | Surface pressure (pressure) | grid:east (10.221 km) 1016, grid:center (25.388 km) 1016, grid:south (27.676 km) 1016, grid:north (42.079 km) 1016, grid:west (49.325 km) 1016 |
| openweather | Air temperature (temperature) | grid:east (10.221 km) 28.94, grid:center (25.388 km) 29.33, grid:south (27.676 km) 28.78, grid:north (42.079 km) 29.16, grid:west (49.325 km) 29.01 |
| openweather | Wind direction (wind_direction) | grid:east (10.221 km) 176.6, grid:center (25.388 km) 166.8, grid:south (27.676 km) 171.5, grid:north (42.079 km) 138.1, grid:west (49.325 km) 174.8 |
| openweather | Wind speed (wind_speed) | grid:east (10.221 km) 3.105, grid:center (25.388 km) 1.992, grid:south (27.676 km) 2.635, grid:north (42.079 km) 1.882, grid:west (49.325 km) 1.972 |
| purpleair | PM2.5 (pm25) | 128007 (1.625 km) 1.195, 161689 (1.727 km) 5.793, 222589 (1.859 km) 6.249, 268165 (2.296 km) 6.965, 222593 (3.655 km) 6.504, 250205 (4.119 km) 4.371, 98633 (4.643 km) 1.442, 244939 (4.684 km) 5.954, +57 more |
| tceq | Carbon monoxide (co) | 482011039 (0.0 km) 0.1609, 482011035 (14.355 km) 0.1365, 482011052 (29.756 km) 0.4344 |
| tceq | Nitrogen dioxide (no2) | 482011039 (0.0 km) 3.528, 482011015 (10.998 km) 4.617, 482011035 (14.355 km) 8.44, 482011050 (14.566 km) 1.461, 482010026 (14.787 km) 5.9, 482010416 (16.162 km) 5.108, 482011052 (29.756 km) 12.11, 480391004 (30.447 km) 3.469, +4 more |
| tceq | Sulfur dioxide (so2) | 482011039 (0.0 km) 0.2806, 482011035 (14.355 km) -0.2016, 482010051 (33.805 km) -0.3742 |


Ground observation locations in the stored 72-hour context
dashed rings are great-circle distance from the anomaly
asos (7)
openaq (96)
purpleair (65)
tceq (13)
anomaly
50 km
40 km
30 km
20 km
10 km
Latitude (degrees north)
Longitude (degrees east)


Event-nearest wind vectors from stored values (arrow points downwind)
asos
longest arrow = 4.63 m/s
noaa_gfs
longest arrow = 4.87 m/s
openweather
longest arrow = 3.41 m/s

## Claims

Mark exactly one box per claim: V (Valid), I (Invalid), or U (Unsure). To skip a claim, leave all three boxes blank; a blank is recorded as missing and never turned into Unsure. Each claim appears exactly once, and the number printed next to it is how I'll refer to that claim if I need to ask you about it.

| Label definitions, from the labeling guide |
|---|
| Valid (V): the claim holds up at the precision it states, and the evidence in the packet supports it. |
| Invalid (I): the evidence in the packet contradicts the claim, the claim misstates a measurement, or the physical reasoning does not hold. |
| Unsure (U): you cannot tell from what is shown. This covers thin evidence, measurements that cannot resolve the claim, ambiguous wording, and anything outside what you are comfortable judging. |
| Missing or thin evidence means Unsure, not Invalid. |
| A cause that is plausible but unproven stays Unsure unless the evidence actually contradicts it. |
| Some claims bundle several statements into one sentence. Judge those by their parts. If every part earns the same verdict, use that verdict. |

### Claim 1 of 36

A short-duration local combustion plume from nearby industrial equipment, flaring, ships, rail, trucks, or other fuel-burning sources is a plausible cause of the CO spike because the anomaly was large at one TCEQ monitor while nearby NO2 and SO2 at the same time remained low, a pattern consistent with a narrow CO-rich plume rather than a broad regional photochemical event.

| Mark exactly one: | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 2 of 36

The TCEQ carbon monoxide anomaly at 29.670, -95.129 had a z-score of 5.91885943430829 and was classified as severe.

| Mark exactly one: | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 3 of 36

The localized ground-level impact at TCEQ monitor 482011039 is supported by substantially lower contemporaneous carbon monoxide levels recorded at nearby TCEQ monitor 482011035.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 4 of 36

A shallow morning boundary layer and light surface winds restricted atmospheric dispersion, trapping surface carbon monoxide emissions near the ground prior to afternoon boundary layer growth.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 5 of 36

The air quality anomaly in Houston, Texas on July 10, 2026 was likely caused by a combination of factors including weather patterns and local emissions.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 6 of 36

The temperature was unusually high on July 10th, 2026, which may have contributed to the air quality anomaly.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

## Claim 7 of 36

Ground monitor 482011039 recorded a peak carbon monoxide concentration of 0.5 ppm at 16:00 UTC on July 10, 2026.

| Mark exactly one: | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

## Claim 8 of 36

Satellite observations from Sentinel-5P detected an elevated total column carbon monoxide density over the broader Houston area on July 10, 2026, supporting a regional enhancement in carbon monoxide.

| Mark exactly one: | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

## Claim 9 of 36

A severe carbon monoxide concentration anomaly was detected at location (29.670, -95.129) on 2026-07-10T16:00:00+00:00 with an observed value of 0.5 ppm compared to an expected mean of approximately 0.156 ppm, resulting in a z-score of 5.92.

| Mark exactly one: | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 10 of 36

A strong wind with a speed of over 3.41 m/s blew through Houston on July 10th, 2026, potentially dispersing pollutants and affecting air quality.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 11 of 36

Regional smoke or area-wide secondary pollution is a less plausible primary cause of the CO spike because PM2.5 and PM10 near the event were only modestly elevated, ozone was not unusually high, and the satellite CO column enhancement was only slight relative to the prior overpass day.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 12 of 36

The air quality anomaly is related to particulate matter (PM2.5) with a value of 7.7 ug/m3.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 13 of 36

The anomalous pollutant was carbon monoxide measured by a TCEQ monitor at 29.670, -95.129 in Houston at 2026-07-10T16:00:00+00:00.

| Mark exactly one: | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

Note (optional):

### Claim 14 of 36

The air quality anomaly in Houston, Texas is caused by high levels of SO2 particles.

| Mark exactly one: | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

Note (optional):

### Claim 15 of 36

The air quality anomaly in Houston, Texas is caused by high levels of PM2.5 particles.

| Mark exactly one: | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

Note (optional):

### Claim 16 of 36

Meteorological transport from sources to the south or southwest of the monitor is a plausible cause of the CO spike because surface winds near the event were from about 161 to 200 degrees with light-to-moderate speeds near 2.6 to 3.4 m/s, which would advect emissions from the Gulf/Ship Channel side toward the site while still allowing elevated near-source concentrations.

| Mark exactly one: | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 17 of 36

ASOS weather observations near the anomaly recorded surface wind speeds of approximately 2.5 m/s from the south-southeast around 16:00 UTC on July 10, 2026.

| Mark exactly one: | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 18 of 36

Surface winds near the anomaly were southerly to southwesterly at roughly 2.6 to 3.4 m/s around 16:00 UTC, which is consistent with transport of a nearby combustion plume into the monitor rather than calm stagnation alone.

| Mark exactly one: | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 19 of 36

The air quality anomaly in Houston, Texas on July 10, 2026 was also characterized by elevated levels of NO2.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 20 of 36

The air quality anomaly in Houston, Texas is caused by high levels of NO2 particles.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 21 of 36

Sentinel-5P atmospheric observations recorded an increase in total column carbon monoxide over the Houston region from 0.0244 mol/m^2 on July 9, 2026, to 0.0263 mol/m^2 on July 10, 2026.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 22 of 36

The TCEQ carbon monoxide concentration at 29.670, -95.129 was 0.5 ppm at 2026-07-10T16:00:00+00:00, compared with an expected value of 0.15572519083969463 ppm.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 23 of 36

The wind direction at the time of the anomaly was approximately 161.0 degrees.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 24 of 36

Sentinel-5P satellite measurements support a broader regional enhancement, recording an increase in total column carbon monoxide from 0.0244 mol/m² on July 9 to 0.0262 mol/m² on July 10.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 25 of 36

The regional-stagnation-or-biomass-smoke hypothesis is contradicted because boundary-layer depth was high by afternoon near 1686 m rather than strongly suppressed, PM2.5 near the site was only about 7.7 to 9.9 ug/m3, PM10 near the site was 24 ug/m3, ozone near the site was 0.023 ppm, and satellite SO2 was not elevated on the same day, so the data do not show a broad smoke or multi-pollutant accumulation episode accompanying the CO spike.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 26 of 36

Industrial combustion or process emissions originating from facilities upwind in the Houston Ship Channel region contributed directly to the elevated surface carbon monoxide concentration recorded at TCEQ monitor 482011039.

| Mark exactly one: | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 27 of 36

The air quality anomaly in Houston, Texas on July 10, 2026 was characterized by elevated levels of PM2.5.

| Mark exactly one: | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 28 of 36

Co-pollutants and particles near the anomaly were not elevated in a way expected for a broad regional smoke or photochemical event, because the coincident site values were $NO_2$ 1.9 ppb and $SO_2$ 0.2 ppb from tceq, ozone 0.023 ppm from openaq, PM2.5 9.9 ug/m3 from openaq, and PM2.5 7.7 ug/m3 from purpleair.

| Mark exactly one: | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 29 of 36

The TCEQ monitor at 29.670, -95.129 measured CO of 0.5 ppm at 2026-07-10 16:00 UTC, far above the site-specific expected value of 0.1557 ppm, indicating a severe localized CO enhancement.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 30 of 36

TCEQ ground monitor 482011039, positioned at the anomaly location (29.670, -95.129), recorded the elevated carbon monoxide reading of 0.5 ppm at 16:00 UTC on July 10, 2026.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 31 of 36

NOAA GFS boundary layer height data supports atmospheric trapping during the morning hours, showing a low mixing depth of 115.4 meters around 12:00 UTC on July 10 before expanding to 1552 meters by 18:00 UTC.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 32 of 36

The PM2.5 levels were higher than usual in the area, possibly due to nearby industrial or agricultural activities, which could have contributed to the air quality anomaly.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 33 of 36

Satellite measurements from Sentinel-5P showed higher total column carbon monoxide levels over the Houston region on July 10, 2026, than on July 9, 2026.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 34 of 36

The local-combustion-plume hypothesis is also supported because co-pollutants associated with fresh industrial or mobile combustion were muted at the anomaly time, with site NO2 at 1.9 ppb and site SO2 at 0.2 ppb, which is consistent with a CO-enhanced plume from a source or plume age that does not strongly elevate NO2 or SO2 at the receptor.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 35 of 36

The local-combustion-plume hypothesis is supported because the TCEQ monitor at the anomaly site recorded CO of 0.5 ppm against a site mean near 0.1609 ppm, while winds near the event were from roughly 161 to 200 degrees at about 2.6 to 3.4 m/s, a pattern consistent with transport from sources located south to south-southwest of the monitor.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

### Claim 36 of 36

The temperature at the time of the anomaly was approximately 30.26°C.

| **Mark exactly one:** | ☐ V | ☐ I | ☐ U |
|---|---|---|---|

**Note (optional):**

## Optional: most likely cause

Your best guess at the true cause of this event, in your own words. "Insufficient evidence to determine a single cause" is a perfectly fine answer.